

007 Memory Return

Return, Lift, Difference, and Recoverable Occurrence

Anthony Vito Coppa

Original construction: 20 July 2026

Canonical standalone edition v0.2: 29 August 2026

Status. Definitions are DECLARED. The algebraic, covering, archive, and non-erasure results proved below are DERIVED under their displayed hypotheses. The standard rotation double cover is used only as an explicit mathematical witness.

Abstract. A return can close at one reader while a lifted endpoint, a representation read, or a retained history remains different. This paper types those offices separately and then states the minimum bridge required to place them in one construction. The bridge is explicit: a declared coupling map sends the returned path to the based loop whose lift carries the endpoint defect. Receipt symbols are separated from receipt spaces and archives; recoverability is stated by a left inverse at a declared resolution. A standard $SU(2) \rightarrow SO(3)$ witness then proves both non-silent and silent return with retained occurrence. The construction is called *Memory Return*; it is not a definition of memory in general.

1 Return at a declared reader

Let X be a state space and let

$$q : X \rightarrow Y$$

be a declared reader.

Definition 1.1 (Reader return). *A path $x : [0, 1] \rightarrow X$ is a q -return when*

$$q(x(1)) = q(x(0)).$$

The equality is in Y only.

Hence reader return does not by itself imply carrier return:

$$q(x(1)) = q(x(0)) \not\Rightarrow x(1) = x(0).$$

The reader fixes what has returned and at what resolution.

2 Representation difference

Let G be a group, let V be a vector space over an arbitrary field $\mathbb{F}$, and let

$$\varrho : G \rightarrow GL(V)$$

be a representation.

Definition 2.1 (Difference observable). *For $g \in G$, define*

$$\Psi_{\varrho}(g) := \varrho(g) - I_V \in \text{End}(V).$$

Theorem 2.2 (Difference composition). *For all $g, h \in G$,*

$$\Psi_\varrho(gh) = \Psi_\varrho(g) + \Psi_\varrho(h) + \Psi_\varrho(g)\Psi_\varrho(h).$$

Equivalently, with

$$A \star B := A + B + AB,$$

one has

$$\Psi_\varrho(gh) = \Psi_\varrho(g) \star \Psi_\varrho(h).$$

Proof. Since ϱ is multiplicative,

$$\varrho(gh) - I_V = (I_V + \Psi_\varrho(g))(I_V + \Psi_\varrho(h)) - I_V.$$

Expanding gives the formula. Moreover,

$$I_V + (A \star B) = (I_V + A)(I_V + B),$$

so associativity of composition implies associativity of $\star$, and 0 is its identity. □

Theorem 2.3 (Silent locus).

$$\Psi_\varrho(g) = 0 \iff g \in \ker \varrho.$$

Hence a zero represented difference forces $g = e$ only when ϱ is faithful.

Proof. $\Psi_\varrho(g) = 0$ is exactly $\varrho(g) = I_V$, which is the definition of membership in $\ker \varrho$. □

3 Lifted endpoint defect

Let

$$p : \tilde{B} \rightarrow B$$

be a connected regular covering. Let its deck group Γ act on $\tilde{B}$, and assume the usual regular-cover property that Γ acts transitively on each fiber. Let $\gamma : [0, 1] \rightarrow B$ be a based loop and $\tilde{\gamma}$ the chosen lift beginning at $\tilde{b}_0$.

Definition 3.1 (Endpoint defect). *The endpoint defect is the unique $\varepsilon \in \Gamma$ satisfying*

$$\tilde{\gamma}(1) = \varepsilon \cdot \tilde{\gamma}(0).$$

Existence follows because the two endpoints lie in the same fiber and the deck action is transitive there; uniqueness follows from freeness of the deck action. Base closure and lifted closure are distinct predicates:

$$\gamma(1) = \gamma(0) \quad \text{while possibly} \quad \tilde{\gamma}(1) \neq \tilde{\gamma}(0).$$

4 Receipt, archive, and resolution-bounded recovery

Let $\mathcal{R}$ be a receipt universe. Define the length- m archive by

$$\text{Archive}_m(\mathcal{R}) := \mathcal{R}^m.$$

For $M \in \text{Archive}_m(\mathcal{R})$ and $r \in \mathcal{R}$, write $\text{snoc}(M, r)$ for append.

Theorem 4.1 (Split append).

$$\text{dropLast}(\text{snoc}(M, r)) = M, \quad \text{last}(\text{snoc}(M, r)) = r.$$

Thus append is injective in its archive argument, and equal payloads appended at different addresses remain distinct occurrences.

Proof. Both identities are the defining projection laws of finite sequence append. If $\text{snoc}(M, r) = \text{snoc}(M', r)$, applying dropLast gives $M = M'$. $\square$

Theorem 4.2 (Persistent indexed retention). *Let S be any later suffix and let*

$$A = M \parallel [r] \parallel S,$$

where M has length m . Then

$$\text{entry}_m(A) = r.$$

Later appends do not erase the earlier receipt.

Proof. Concatenation preserves the first $m + 1$ addresses. The element appended immediately after the length- m prefix therefore remains at address m under zero-based indexing. $\square$

Fix a promised history resolution δ . Let W_δ be the admitted history classes at that resolution and let $\mathcal{R}_\delta$ be the corresponding receipt-code space.

Definition 4.3 (Lossless ingress at resolution δ). *A lossless ingress pair is*

$$\text{Enc}_\delta : W_\delta \rightarrow \mathcal{R}_\delta, \quad \text{Dec}_\delta : \mathcal{R}_\delta \rightarrow W_\delta$$

with

$$\text{Dec}_\delta \circ \text{Enc}_\delta = \text{id}_{W_\delta}.$$

The decoder may be partial outside the image of the encoder.

Theorem 4.4 (Archive-ingress composition). *Fix a prefix archive M of length m and define*

$$A_M(w) := M \parallel [\text{Enc}_\delta(w)].$$

After any later suffix S is appended,

$$\text{Dec}_\delta(\text{entry}_m(A_M(w) \parallel S)) = w.$$

Hence history-to-receipt-to-archive ingress is split at the declared resolution.

Proof. Theorem 4.2 gives

$$\text{entry}_m(A_M(w) \parallel S) = \text{Enc}_\delta(w).$$

Apply Dec_δ and use Definition 4.3. $\square$

The theorem recovers exactly the distinctions represented by W_δ and encoded by Enc_δ . It cannot recover a distinction that was never encoded.

5 The Memory Return datum

The return path and the lifted loop must be connected by an explicit map; co-residence in a tuple is not enough.

Definition 5.1 (Memory Return datum). *A Memory Return datum at resolution δ is the four-office record*

$$\boxed{\text{MR}_\delta = (\text{Ret}, \text{Lift}, \text{Diff}, \text{Rec})}$$

with

$$\text{Ret} = (q, x),$$

where $x : [0, 1] \rightarrow X$ is a q -return;

$$\text{Lift} = (\alpha, p, \gamma, \tilde{\gamma}, \varepsilon),$$

where $\alpha : X \rightarrow B$ is a declared continuous coupling map satisfying

$$\boxed{\gamma = \alpha \circ x}$$

and γ is a based loop, $p : \tilde{B} \rightarrow B$ is the declared regular cover, $\tilde{\gamma}$ is the chosen lift, and ε is its endpoint defect;

$$\text{Diff} = (\varrho, \Psi_\varrho(\varepsilon)),$$

where $\varrho : \Gamma \rightarrow GL(V)$ is a declared representation; and

$$\text{Rec} = (w, r, m), \quad r = \text{Enc}_\delta(w) \in \mathcal{R}_\delta,$$

where $w \in W_\delta$ is the admitted traversal-history class and r is retained at archive address m .

The coupling equation $\gamma = \alpha \circ x$ is load-bearing. It is the typed bridge that the earlier phrase “action witness” left unspecified. Without such a map or an equivalent explicit relation, a reader-return path and an unrelated lifted loop do not constitute one Memory Return datum.

No equality of numerical representatives licenses type collapse. A scalar -1 , a central element $-I$, and an operator $-I_V$ may correspond under declared maps; they are not one object.

6 Standard double-cover witness

Take

$$B = X = SO(3), \quad q = \text{id}_{SO(3)}, \quad \alpha = \text{id}_{SO(3)},$$

and use the standard connected regular cover

$$p : SU(2) \rightarrow SO(3)$$

with deck group $\Gamma = \{\pm I_2\}$. Let $\varrho : \Gamma \rightarrow GL_2(\mathbb{C})$ be the defining inclusion.

For a generator loop corresponding to a 2π rotation, the chosen lift beginning at I_2 ends at $-I_2$. Thus

$$\varepsilon_{2\pi} = -I_2, \quad \Psi_\varrho(\varepsilon_{2\pi}) = -2I_2.$$

The base loop is a q -return because it is based, while the lifted endpoint and represented difference remain nontrivial.

For the concatenated 4π loop,

$$\varepsilon_{4\pi} = (-I_2)^2 = I_2.$$

Using Theorem 2.2,

$$\Psi_{4\pi} = (-2I_2) \star (-2I_2) = -4I_2 + 4I_2 = 0.$$

The same result follows from Theorem 2.3 because $\varepsilon_{4\pi} = e$.

If W_δ distinguishes either traversed loop from the null history and the receipt ingress is lossless, Theorem 4.4 preserves that distinction after later appends.

7 Non-erasure and silence

Theorem 7.1 (Memory Return non-erasure). *Within the class of Memory Return data, reader closure does not determine lifted defect, represented difference, or traversal occurrence. In particular, none of the following implications is valid without an additional hypothesis:*

$$q(x(1)) = q(x(0)) \implies \varepsilon = e,$$

$$q(x(1)) = q(x(0)) \implies \Psi_\varrho(\varepsilon) = 0,$$

$$q(x(1)) = q(x(0)) \implies w = w_\varnothing.$$

Proof. The 2π double-cover witness above is a reader return with $\varepsilon = -I_2 \neq e$ and $\Psi_\varrho(\varepsilon) = -2I_2 \neq 0$. Hence the first two conclusions do not follow from reader closure. The 4π witness is a reader return with represented difference 0 while, at any history resolution that distinguishes the traversed loop from the null history, lossless ingress gives a receipt recovering $w_{4\pi} \neq w_\varnothing$. Hence reader closure, including represented silence, does not imply empty occurrence. $\square$

Definition 7.2 (Silent Memory Return). *A Silent Memory Return is a Memory Return datum satisfying*

$$q(x(1)) = q(x(0)), \quad \Psi_\varrho(\varepsilon) = 0,$$

while its adequate receipt distinguishes the admitted traversal history from the null history:

$$w \neq w_\varnothing.$$

Silence here means closure at the selected representation/read, not absence of occurrence.

8 Office separation

Office	Carrier	Question
Return	reader codomain Y	did the declared read close?
Lift	cover / deck group Γ	what finer endpoint distinction remains?
Difference	$\text{End}(V)$	what does the representation distinguish?
Receipt	$W_\delta, \mathcal{R}_\delta$, archive address	which occurrence can be recovered?

These offices are coupled only by the maps explicitly declared in Definition 5.1. None may be silently substituted for another.

9 Boundary

This paper establishes:

- typed return relative to a reader;
- the exact representation-difference composition law and silent locus;
- lifted endpoint defect under a declared regular cover;
- split append and persistent indexed retention;
- left-invertible resolution-bounded ingress;
- an explicit coupling map joining a returned path to the lifted loop used by the Memory Return datum;
- concrete $2\pi/4\pi$ witnesses showing that reader closure, lifted defect, represented difference, and occurrence are different predicates;
- Silent Memory Return as a defined special case of represented closure with non-null recoverable occurrence.

It does not establish:

- that all memory is return;
- that storage requires recurrence, curvature, or holonomy;
- that a deck element contains semantic content;
- that equal numerical representatives identify independently typed objects;
- that a return architecture is automatically a physical, biological, or cognitive mechanism;
- that endpoint recurrence supplies a Berry phase without the required native transport data.

Source record

The following entries are provenance or standard mathematical background only. Every theorem needed by the present Language Mechanics construction is stated and proved locally above.

- A. V. Coppa, *Memory Return: Observable Difference, Central Lift, Lossless Bookkeeping*, Draft 0.1, 20 July 2026.
- A. V. Coppa, *Memory Return Closure Certificate: Persistent Lossless Ingress and an Exact Paraxial Realization of Q^2* , 21 August 2026. Only the receipt/ingress construction is retained here; the optical realization remains outside this form.
- A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002, Chapter 1, for standard covering-space background.